

SOLARSHARE · P2P ENERGY EXCHANGE

Trade the sun next door.

Peer-to-peer rooftop solar for Indian households — priced in ₹, settled on real Postgres, and stamped into a SHA-256 hash-chained ledger you can verify in one click.

₹8.50

market base price / kWh

105

unit tests passing

1 click

verify chain from genesis

6 slides · ~6 minutes · live demo of the running app

BUILT BY

Jeet Patel

ledger + trading engine

Shlok Vaghasiya

marketplace UI + deal flow

Dhruvit Vadukiya

data layer, tests, demo

Live web app — real accounts, real SQL, real hash chain. Optional read-only bridge: Polygon Amoy testnet (chainId 80002)



₹500

welcome credit on sign-up

3.0 kWp

simulated rooftop per account

₹5.5-13.5

live band, Feeder 14

THE PROBLEM

Rooftop solar has no market — only a meter

You generate 3 kW on your roof and ₹0 on your terms: surplus leaves at a fixed feed-in rate while the flat next door buys it back at retail.

01

Locked surplus

Net metering exports your daytime over-generation at a fixed feed-in tariff, settled monthly. You choose no price, no buyer, no timing — while your evening peak is billed at full retail.

Sell into a curve, not a market.

02

Neighbours never meet

A producer and a buyer on the same feeder, 400 m apart, cannot trade: no shared order book, no per-peer identity, no settlement rail, no record of who owed whom.

Same wire. No meeting point.

03

'Blockchain energy' that isn't

Most demo projects ship a mockup and a logo. Judges and utilities ask one question — where is the record? A ledger you cannot re-hash from raw rows is a slide, not a system.

Trust needs proof.

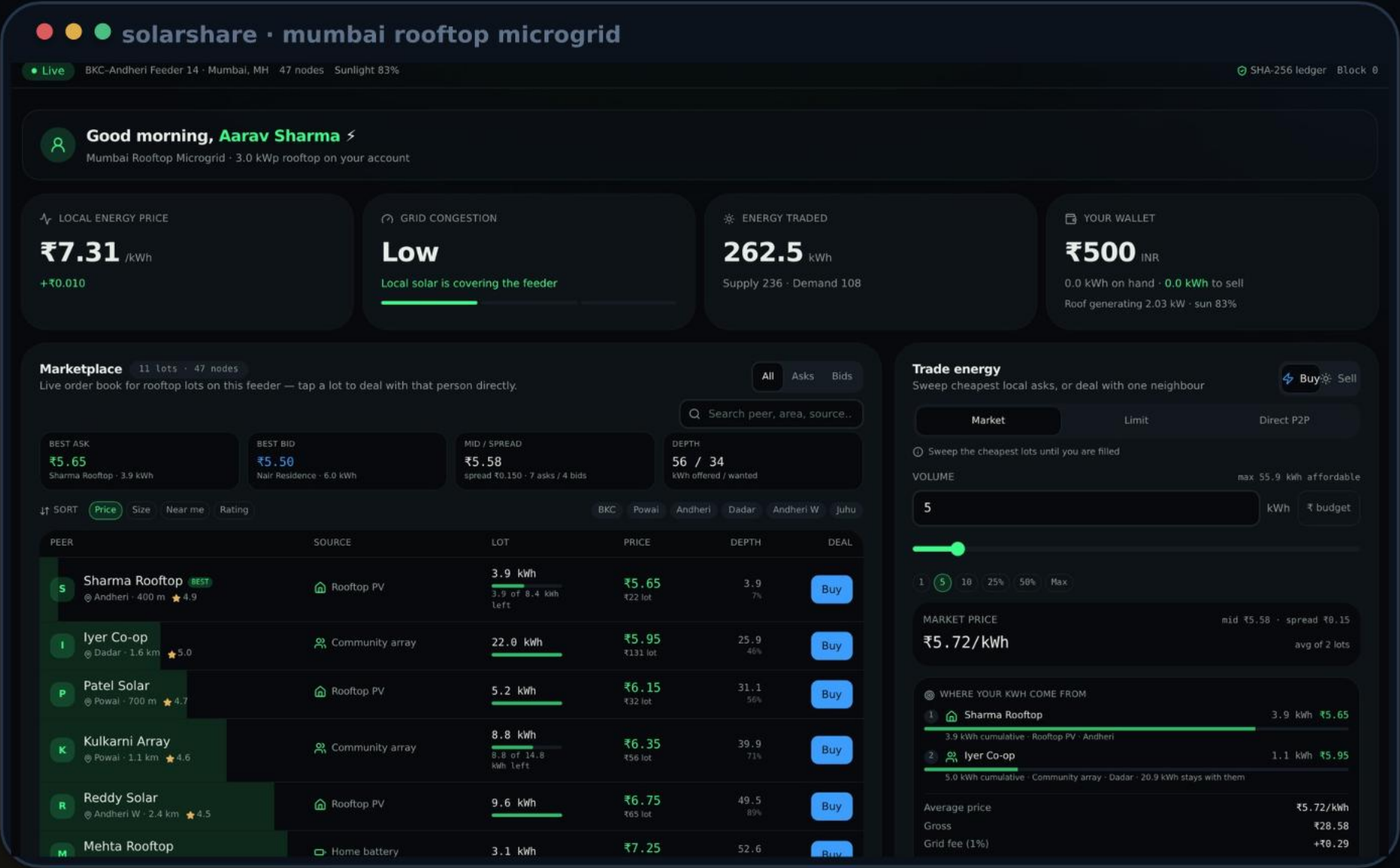
WHY NOW

- Residential rooftop is the fastest-growing solar segment
- Retail solar P2P value trades roughly ₹4-₹12 / kWh
- Every kWh traded peer-to-peer keeps that margin inside the feeder

THE PRODUCT

One wallet. Two directions. Zero roles.

Every account can Buy kWh and Sell kWh — ₹ balances, a real order book, a live route preview. This is the running app, not a mockup.



Live order book — peer, area, distance, reliability, lot left · best ask ₹5.65 · Trade panel with the complete route before you sign.

SAME WALLET, BOTH SIDES

- ₹ first, always
Prices, balances and ledger totals are INR. Base ₹8.5/kWh, live band ₹5.5-13.5, IST solar curve for Mumbai, BKC-Andheri Feeder 14.
- Sign up → trade in 30 s
Email + password lands in real SQL tables. New profiles get ₹500 credit and a 3 kWp rooftop that keeps generating through the day.
- No producer/consumer tag
Surplus accumulates from the panel curve and can be sold into the book the same minute you need to buy.
- UPI rails for money in and out
Top-up and withdraw against stored payment methods; each movement is a ₹ row plus a minted block.

THE 60-SECOND DEMO

sign up → ₹500 → pick a lot → sign → wallet and chain head move together

HOW TRADING WORKS

Three ways to trade — all quoted before you sign

src/lib/deal.ts is a pure TypeScript engine: no React, no network, no DB. The same maths drives the preview and the settlement.

solarshare · mumbai rooftop microgrid

Direct P2P deal

You buy

×

Pick how much of Sharma Rooftop's lot to take — a slice or the whole lot — and optionally spread delivery over instalments. Every instalment settles atomically and mints its own block.

S

Sharma Rooftop

⌂ Andheri · 400 m ⌂ Rooftop PV

★ 4.9 · 212 settled 0x3A91...B0E1

THEIR PRICE

₹5.65/kWh

3.9 kWh left

VOLUME FROM THIS LOT

₹ budget

2

kWh

Min 0.5

25%

50%

75%

Full lot

2.0 of 8.4 kWh left

Takes 24% of their lot · 1.9 kWh stays with Sharma

PRICE

⚙ Accept theirs

Settling at their listed ₹5.65/kWh. Switch to "Your offer" to negotiate inside a 2.5% band.

DELIVERY

One delivery

2 instalments

3 instalments

4 instalments

One atomic settlement, one block.

Volume

Price

Gross

2.0 kWh

₹5.65/kWh

₹11.30

Direct P2P deal sheet: a slice of one neighbour's lot, offer band, instalments, margin vs mid, blocks to mint.

MARKET SWEEP

routeFill()

Walk the cheapest asks (or richest bids) until you are filled — across as many neighbours as it takes. One order can produce several legs, and one leg can take part of a lot.

LIMIT ORDER

rests on the book

Set your own price with an explicit margin versus the mid. The unfilled remainder stays on the order book as your order; a match settles it atomically.

DIRECT P2P DEAL

directDeal() · splitInstalments()

One named neighbour, any slice from their minimum up to the whole lot, an offer inside a ±2.5% band (accepted or countered), and delivery split into instalments — each one its own block.

EVERY QUOTE SHOWS BEFORE YOU SIGN

average price

gross

grid fee 1%

you pay

slippage vs best

margin vs mid

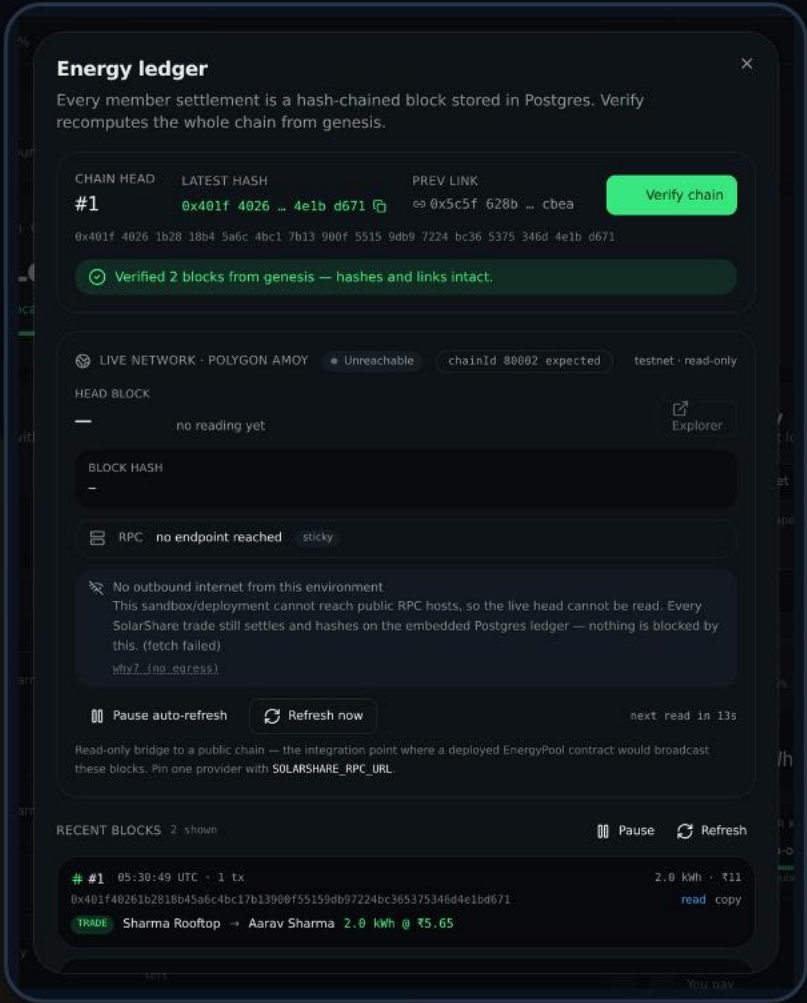
blocks minted

DATA FORGE · SolarShare — peer-to-peer solar exchange, priced in ₹

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Every trade mints a block — and you can re-hash the chain

Not a whitepaper promise: a SHA-256 hash chain stored in the same Postgres as your balance.



Energy ledger dialog: chain head, verification banner, recent blocks.

What a block is

Block n carries the SHA-256 of its canonical payload — block number, previous hash, timestamp, transactions — and links to block n–1. Genesis is deterministic.

What mints one

Top-up, buy, sell, order listing, withdrawal. A 3-instalment deal is 3 blocks, so 'first some kWh, then the next some' is three auditable records, not one.

verifyChain()

Recomputes every hash from raw DB rows and reports the first bad block: edited transaction → hash mismatch, deleted block → broken link, changed prev hash → broken link.

Live, no reloads

ledger-feed.ts folds each poll into the list: unchanged blocks keep object identity so rows never re-render, new blocks queue and land one at a time, inserts above you are scroll-anchored, polling pauses while the tab hides.

A real chain, read live

JSON-RPC to public Polygon Amoy endpoints, chainId verified before a reading is accepted, last healthy endpoint kept sticky, and an honest reason shown when the sandbox has no egress.

Shipped today — and defensible in Q&A

We would rather hand you the exact boundary of what is simulated than be caught inflating it.

STACK

Frontend	React 19 · TanStack Start + Router with SSR · Tailwind 4 · Recharts · Radix
Server	createServerFn + bearer-token middleware → SQL in the same request path
Auth	better-auth email+password → user/session/account, bcrypt-scrypt, httpOnly cookies
Data	PGLite (embedded PostgreSQL 16) or Neon via DATABASE_URL · migrations 0001-0003
Ledger	ledger-core.ts (pure hashing) · ledger.server.ts (mint/verify) · ledger-feed.ts (live UI)
Chain read	testnet.server.ts → Polygon Amoy, read-only, chainId-guarded, graceful offline

PROOF ON THE RECORD

105 / 105	app tests pass — ledger core, live feed, trading engine, network bridge
6 + 14	tests on the chain (deterministic hashing, genesis, tamper) and on the live ledger list
1 E2E	HTTP smoke mirroring the browser: sign-up → ₹500 → top-up → buy → list → verify → withdraw
SSR	dashboard renders server-side — ₹ balances, market, wallet, all in the HTML
HONEST LIMITS	PGLite is in-memory per process (swap in Postgres for durability) · neighbour lots and ambient activity are simulated client-side · the chain is PoA-shaped, not a consensus network · live Amoy reads depend on egress

NEXT, IN ORDER

1 · Persist	managed Postgres + real UPI payout and settlement records	2 · Share	server-side order book so neighbours are real members, with disputes	3 · Anchor	deploy EnergyPool on Amoy; anchor each block head hash on-chain	4 · Comply	feeder-level P2P within net-metering rules · WhatsApp / UPI-lite reach
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Thank you — DATA FORGE · Jeet Patel · Shlok Vaghasiya · Dhruvit Vadukiya